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Lecture 1 — Introduction to Signal Processing Lecture Notes

ADS203TC Digital Signal Processing
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Part 1: Signals, Systems, and Classification

What is a Signal?

A **signal** is any physical quantity that varies with time, space, or another independent variable, formally modeled as a mathematical function. Signals carry information about the state or behavior of a physical system.

Continuous-time signal $x(t)$: Defined for all $t \in \mathbb{R}$.

Discrete-time signal $x[n]$: Defined only at integer indices $n \in \mathbb{Z}$.

Continuous-valued: Amplitude takes any real value.

Discrete-valued (Quantized): Amplitude restricted to a finite set of levels.

Digital signal: Discrete in **both** time and amplitude.

Signals are further classified as:

- **Deterministic**: Exactly described by a mathematical expression (e.g., $x(t) = 5 \sin(20\pi t)$).
- **Random**: Cannot be precisely predicted; described statistically (e.g., thermal noise).
- **Multichannel**: A vector of signals from multiple sensors, $\mathbf{S}(t) = [s_1(t), s_2(t), s_3(t)]^T$.

The DSP System

A digital signal processing system converts an analog input to a digital sequence, processes it mathematically, and converts the result back to analog:



DSP offers **programmability** (reconfigure via software), **precision** (no component drift), **storage** without degradation, and the ability to implement **complex algorithms** impossible in analog hardware. Its main limitations are bandwidth (constrained by A/D converter speed) and finite word-length effects (quantization noise).

A System and its Properties

A discrete-time **system** is an operator $\mathcal{T}[\cdot]$ mapping an input sequence $x[n]$ to an output sequence $y[n]$:

$$y[n] = \mathcal{T}[x[n]]$$

Systems are characterized by four fundamental properties:

- 1. Linearity:** A system is linear if superposition holds. For any constants a, b and inputs $x_1[n], x_2[n]$:

$$\mathcal{T}[ax_1[n] + bx_2[n]] = a\mathcal{T}[x_1[n]] + b\mathcal{T}[x_2[n]]$$

Example: $y[n] = 3x[n]$ is linear; $y[n] = x^2[n]$ is nonlinear.

- 2. Time-Invariance:** A time shift in the input causes an identical shift in the output:

$$\mathcal{T}[x[n - k]] = y[n - k]$$

Example: $y[n] = x[n] - x[n - 1]$ is time-invariant; $y[n] = n \cdot x[n]$ is time-variant.

- 3. Causality:** The output depends only on present and past inputs, not future ones. *Example:* $y[n] = x[n] + x[n - 1]$ is causal; $y[n] = x[n + 1]$ is non-causal.

- 4. BIBO Stability:** Every bounded input produces a bounded output. If $|x[n]| \leq M_x < \infty$ for all n , then $|y[n]| \leq M_y < \infty$ for all n .

Part 2: Sinusoids, Energy, and Discrete-Time Behavior

The Continuous-Time Sinusoid

The fundamental building block of analog signal analysis:

$$x_a(t) = A \cos(\Omega t + \theta) = A \cos(2\pi F t + \theta)$$

where A is amplitude, Ω is angular frequency (rad/s), F is frequency (Hz), and θ is phase (rad). Key properties:

1. **Universally periodic** with period $T_p = 2\pi/\Omega$ for any Ω .
2. **Unique:** distinct frequencies $\Omega_1 \neq \Omega_2$ produce distinct signals.
3. **Monotonic oscillation rate:** increasing Ω strictly increases the rate of oscillation.

Using Euler's identity $e^{j\phi} = \cos \phi + j \sin \phi$, a sinusoid can be expressed in complex exponential form:

$$A \cos(\Omega t + \theta) = \Re\{A e^{j(\Omega t + \theta)}\} = \frac{A}{2} e^{j\theta} e^{j\Omega t} + \frac{A}{2} e^{-j\theta} e^{-j\Omega t}$$

The complex exponential $e^{j(\Omega t + \theta)}$ represents a rotating phasor in the complex plane; its projection onto the real axis yields the physical sinusoid.

The Discrete-Time Sinusoid — Three Critical Differences

A discrete-time sinusoid is defined at integer indices n :

$$x[n] = A \cos(\omega n + \theta), \quad \omega = 2\pi f$$

where ω is the discrete angular frequency (rad/sample) and f is the discrete frequency (cycles/sample).

Discrete-time sinusoids behave fundamentally differently from their continuous-time counterparts:

Difference 1: Periodicity is not guaranteed. A discrete sinusoid is periodic only if its digital frequency f is a rational number: $f = k/N$ where k, N are integers. For example, $x[n] = \cos(3n)$ is aperiodic because $f = 3/(2\pi)$ is irrational.

Difference 2: Frequencies are not unique (Aliasing). Discrete frequencies separated by integer multiples of 2π are identical:

$$\cos((\omega + 2\pi)n + \theta) = \cos(\omega n + \theta)$$

The **fundamental range** is $-\pi \leq \omega \leq \pi$ (or $0 \leq \omega \leq 2\pi$). Frequencies outside this range are aliases of frequencies inside it.

Difference 3: Oscillation rate is non-monotonic. The highest possible oscillation rate occurs at $\omega = \pi$ (or $f = 1/2$). As ω increases from π to 2π , the oscillation rate *decreases*, behaving like a negative frequency from $-\pi$ to 0.

Energy and Power Signals

For a discrete-time signal $x[n]$, the total energy E and average power P are:

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2, \quad P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

- **Energy signal:** $0 < E < \infty$ (hence $P = 0$). Typical of transient, finite-duration signals (e.g., a rectangular pulse, $x[n] = a^n u[n]$ with $|a| < 1$).
- **Power signal:** $0 < P < \infty$ (hence $E = \infty$). Typical of periodic or stationary random signals (e.g., $x[n] = A \cos(\omega n)$, the unit step $u[n]$).

For a **periodic** signal with fundamental period N , the average power simplifies to computation over one period:

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2$$

Summary

- A **digital signal** is discrete in both time and amplitude. DSP systems convert, process, and reconstruct signals with flexibility and precision.
- **Systems** are classified by linearity, time-invariance, causality, and BIBO stability — these properties determine how a system can be analyzed and designed.
- **Discrete-time sinusoids** exhibit three fundamental differences from continuous-time: conditional periodicity, frequency aliasing, and a non-monotonic oscillation rate capped at $\omega = \pi$.
- Signals are categorized by their **energy** (finite-energy transients) or **average power** (finite-power periodic/random signals).